

MIKU: Interaction-Aware Spatiotemporal Homotopy Corridors for Decoupled Path–Velocity Planning

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Abstract—Path–velocity decomposition supports reliable real-time autonomous-driving planners through mature convex optimization, yet its one-way interface commits to a spatial passing side before resolving temporal interaction. This can discard feasible solutions in dense multi-obstacle and dynamic-crossing scenarios. We present MIKU, an interaction-aware spatiotemporal homotopy-corridor framework that retains the existing path and speed quadratic programs. MIKU builds threat-adaptive margins and bounded-error occupancy tubes, reduces the 2^k passing-side assignments of a conflict group to $k + 1$ continuous lateral bands, and couples Top- K spatial homotopies with a causal multi-conflict temporal graph. Pass-before and yield-after decisions are compiled into bidirectional station–time bounds. Once a discrete homotopy is fixed, both continuous subproblems remain convex. On-demand alternating refinement updates arrival times from the speed solution, while semantic persistence and pass-before commitment stabilize receding-horizon decisions. Across 3500 paired randomized cases in seven scenario families, MIKU achieves a 77.5% collision-free arrival rate, improving over the time-blind baseline by 14.8 percentage points (95% CI: 13.5–16.0), while reducing the collision rate from 2.2% to 1.1%. Its complete-planning P95 latency is 55.93 ms. In 700 receding-horizon episodes, MIKU retains a 10.0-point arrival-rate gain without increasing collisions.

Index Terms—Autonomous driving, motion planning, path–velocity decomposition, spatiotemporal homotopy, safety corridor.

I. INTRODUCTION

PATH–VELOCITY decomposition (PVD) solves lateral geometry first and longitudinal timing second. Since the seminal formulation in [1], this structure has underpinned Frenet, lattice, and optimization-based planners [2], [3], [4], [5]. Apollo EM and its path/speed quadratic programs (QPs) are representative production-oriented implementations [6], [7], [8], [9]. Decomposition makes a high-dimensional problem tractable, but also creates a one-way information bottleneck: the path stage decides *where* to pass, and the speed stage subsequently decides *when* to pass on that fixed path.

This bottleneck becomes restrictive when several obstacles jointly induce lateral and temporal choices. First, obstacle-wise greedy nudging retains only one of several feasible spatial homotopies. Second, the same path can admit pass-before, yield-after, or mixed orders over multiple dynamic conflicts, whereas a one-sided stop bound expresses only yielding. A planner may therefore stop despite an available lateral gap,

or lose a feasible temporal order after prematurely fixing the geometry.

Joint spatiotemporal planners optimize in (s, l, t) or a higher-dimensional state space [10], [11], [12], [13], [14], [15]. They represent coupling directly but generally replace the mature path–speed solver chain. Iterative PVD, candidate pairing, and safe-corridor methods retain more modularity [16], [17], [18], [19], [20], [21], yet still require a compact interface between discrete homotopy decisions and continuous optimization. Grid, hybrid-A*, and mixed-integer approaches can encode passing sides [22], [23], [24], but their discrete combinations grow rapidly. Pedestrian-interaction studies further demonstrate that lateral passing and longitudinal timing should be coordinated [25], [26], [27]. End-to-end systems such as UniAD and VAD learn a unified representation [28], [29], but do not preserve an interpretable convex-QP interface.

We propose MIKU (Multi-obstacle Interaction-aware Kinematic-constraint Unification), a spatiotemporal homotopy-corridor framework for decoupled planning. MIKU searches a small set of spatial and temporal homotopy classes over robust predicted occupancy, then compiles the selected class into box constraints accepted directly by the existing path and speed QPs. The discrete layer answers *where and when*; the continuous layer retains efficient smooth-trajectory optimization.

The contributions are:

- 1) A QP-compatible spatiotemporal homotopy framework that unifies robust occupancy, Top- K spatial alternatives, multi-conflict temporal ordering, and bidirectional ST constraints without replacing the Apollo-style continuous solvers.
- 2) A continuous-partition result showing that the best of 2^k passing-side assignments for a lateral conflict group is covered by $k + 1$ bands. This yields an exact $O(k \log k)$ maximum-gap scan and an exact K -best layered spatial search.
- 3) A causal temporal-homotopy graph whose pass-before, yield-after, and intermediate safe-window choices are converted into lower and upper ST bounds, with ranked QP fallback.
- 4) Bounded-error occupancy tubes, envelope-aware ST mapping, on-demand path–speed refinement, and receding-horizon semantic commitment, while preserving convexity for every fixed homotopy.
- 5) Paired evaluation on 3500 randomized cases and 700 receding-horizon episodes, complemented by seven

TABLE I
STRUCTURAL COMPARISON WITH REPRESENTATIVE PLANNING
FAMILIES

Method family	Spatial choice	Temporal order	Retains P/S QPs
Apollo EM/DL-IAPS [6], [9]	Heuristic	ST decision	Yes
Joint grid/NMPC [11], [14]	Joint search	Joint search	No
Pairing/iteration [17], [18]	Samples	Iteration	Partial
ST corridors [19], [20]	Corridor	Explicit time	No
MIKU	$k + 1$, Top- K	Causal graph	Yes

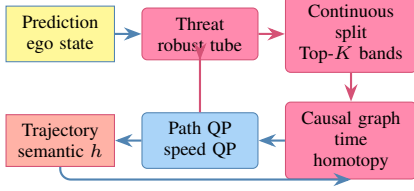


Fig. 1. MIKU wraps discrete homotopy generation and validation around the existing path and speed QPs.

large-scale ablations and a joint spatiotemporal reference.

II. PROBLEM RESTATEMENT

In Frenet coordinates, a trajectory is $\xi(t) = (s(t), l(s(t)))$. Let the physical road boundaries be $[b_{\text{road}}^-(s), b_{\text{road}}^+(s)]$ and the ego width be W_e . The center-feasible road interval is

$$\mathcal{B}(s) = \left[b_{\text{road}}^-(s) + \frac{W_e}{2} + d_{\text{road}}, b_{\text{road}}^+(s) - \frac{W_e}{2} - d_{\text{road}} \right]. \quad (1)$$

Obstacle i has predicted occupancy $O_i(t) = [s_i^-(t), s_i^+(t)] \times [l_i^-(t), l_i^+(t)]$. The path QP optimizes lateral samples l_m at stations s_m ; the speed QP optimizes s_j at times t_j . Standard PVD is

$$l^* = \arg \min_{l \in \mathcal{F}_l} J_l(l), \quad s^* = \arg \min_{s \in \mathcal{F}_s(l^*)} J_s(s). \quad (2)$$

Because $\mathcal{F}_s$ is conditioned on one fixed path, the speed stage cannot recover a discarded spatial class. Likewise, an ST mapper that creates only an upper stop boundary cannot express pass-before behavior.

We restate the task as selecting a discrete label

$$h = (h_l, h_t) \in \mathcal{H}_l \times \mathcal{H}_t \quad (3)$$

and compiling it into lateral and temporal corridors $\mathcal{C}_l(h_l)$ and $\mathcal{C}_t(h_t | l)$ such that

$$l(s) \in \mathcal{C}_l(h_l), \quad s(t) \in \mathcal{C}_t(h_t | l), \quad \xi(t) \notin O_i^{\text{rob}}(t), \quad \forall i, t. \quad (4)$$

A spatial label denotes on which side of the trajectory an obstacle lies. A temporal label denotes whether the ego passes a conflict before or after its robust occupancy. MIKU searches labels in a finite graph and optimizes trajectory quality only after a label is fixed.

III. MIKU SPATIOTEMPORAL HOMOTOPY CORRIDORS

Fig. 1 summarizes the pipeline. Apollo-style boundary construction, path optimization, ST mapping, and speed optimization provide the continuous backbone. MIKU contributes

the robust constraint representation, spatial/temporal candidate organization, bidirectional information transfer, and receding-horizon consistency.

A. Interaction-Aware Robust Occupancy

We combine collision imminence, longitudinal overlap, relative velocity, obstacle type, and local interaction density in a normalized threat score

$$\Theta_i = \sum_{r=1}^5 w_r f_r(i), \quad w_r \geq 0, \quad \sum_r w_r = 1. \quad (5)$$

The lateral expansion for obstacle i is

$$\delta_i = \frac{W_e}{2} + d_0 + d_\Theta \Theta_i. \quad (6)$$

Equation (1) contracts the physical road into a center-feasible interval, while (6) expands each obstacle into a center-forbidden interval; the ego width is therefore counted once at each physical boundary rather than subtracted again from the remaining gap.

For dynamic obstacle i , bounded longitudinal and lateral prediction errors grow as

$$\rho_{q,i}(t) = \rho_{q,i}^0 + t \dot{\rho}_{q,i}, \quad q \in \{s, l\}. \quad (7)$$

The robust tube is

$$O_i^{\text{rob}}(t) = O_i(t) \oplus [-\rho_{s,i}(t), \rho_{s,i}(t)] \times [-\rho_{l,i}(t), \rho_{l,i}(t)]. \quad (8)$$

The same tube is used by lateral projection, ST mapping, and planning-time safety validation. During ST mapping, MIKU checks the complete lateral envelope of the curved path over the vehicle-obstacle longitudinal Minkowski interval, rather than querying $l(s)$ only at the obstacle center. This prevents missed corner overlap.

B. Continuous Partitions and K-Best Spatial Homotopies

An initial arrival map $\tau(s)$ queries (8) and produces center-forbidden intervals $[u_i(s), v_i(s)]$. A longitudinal sweep divides overlapping obstacles into conflict groups. For a group of k obstacles, sort by nondecreasing u_i and define, for $p = 0, \dots, k$,

$$V_p = \max\{l_{\text{road}}^-, v_{(1)}, \dots, v_{(p)}\}, \quad (9)$$

$$U_p = \min\{l_{\text{road}}^+, u_{(p+1)}\}, \quad g_p = U_p - V_p, \quad (10)$$

where empty prefixes and suffixes use the corresponding road bound. Split p places the first p obstacles on one side of the band $[V_p, U_p]$ and the rest on the other.

Lemma 1 (Continuous partition). *For every feasible binary passing-side assignment, there exists a noninterleaved assignment with no smaller center-feasible width: a prefix of the ordered obstacles lies on one side and the remaining suffix lies on the other.*

Proof. If an assignment contains an inversion in the u order, removing that inversion neither decreases the active upper-bound minimum nor increases the active lower-bound maximum. Repeating the exchange yields one split point without reducing width. $\square$

Theorem 1 (Exact maximum gap). *The maximum center-feasible width among all 2^k passing-side assignments equals*

$$g^* = \max_{p \in \{0, \dots, k\}} (U_p - V_p). \quad (11)$$

It is obtained in $O(k \log k)$ time by sorting, a prefix-maximum scan, and a linear maximum search.

Proof. For split p , intersection of all lower-side constraints gives V_p , and intersection of all upper-side constraints gives U_p . Thus its exact width is g_p . Lemma 1 guarantees that at least one global optimum is among these $k + 1$ splits. $\square$

MIKU ranks every positive-width $[V_p, U_p]$ using width, reference-line displacement, and inter-group center transition. To obtain global alternatives, a layered dynamic program retains the K best prefixes ending at each current band. Future cost depends on a prefix only through its terminal band, so this is an exact K-best procedure for the layered first-order graph. With M groups and at most Q retained bands per group, its complexity is $O(MKQ^2 \log(KQ))$. We use $Q = K = 3$. Ranked spatial candidates are sent to the QPs until one is certified feasible.

C. Causal Temporal Homotopies and Bidirectional ST Bounds

For a spatial candidate, intersection with robust tubes yields station-ordered conflicts $c_r = (s_r, \mathcal{I}_r)$, where $\mathcal{I}_r$ is a union of occupied time intervals. Connected components of the complement are safe-window nodes. A window before the first occupied interval is labeled *pass-before*; a window after the last interval is *yield-after*; an intermediate window combines yielding to one occupancy and passing before the next.

Two nodes at successive conflict stations are connected only if their candidate arrival times obey

$$t_b - t_a \geq \frac{s_b - s_a}{v_{\max}}. \quad (12)$$

The graph cost combines deviation from nominal arrival, waiting, class switching, and receding-horizon persistence. A deterministic beam of width eight returns ranked complete homotopies. For an occupied interval $[t_i^-, t_i^+]$ and longitudinal collision range $[s_i^-, s_i^+]$, the selected class is compiled as

$$\text{pass-before: } s(t_j) \geq s_i^+ + \varepsilon_s, \quad t_j \geq t_i^+, \quad (13)$$

$$\text{yield-after: } s(t_j) \leq s_i^- - \varepsilon_s, \quad t_j < t_i^-. \quad (14)$$

Equation (13) is an ST lower bound that commits the ego beyond the conflict after the window closes; (14) is an upper bound that keeps it before the conflict until the window opens. MIKU validates up to three ranked temporal plans for each spatial plan and stops at the first feasible speed QP.

Proposition 1 (Convexity preservation). *If the original path and speed objectives are convex quadratic and their dynamics are linear, fixing (h_l, h_t) leaves both subproblems convex QPs.*

Proof. Spatial bands and (13)–(14) are sample-wise affine lower/upper bounds. Their intersection with the original linear constraints is convex and does not change the quadratic Hessian. $\square$

Input: Ego state, road bounds, predicted obstacles, previous labels
Output: Path $l(s)$, speed $s(t)$, homotopy h
 Build threat margins and robust occupancy tubes; initialize $\tau^{(0)}$;
for $\tau = 0, 1$ **do**
 Generate exact K-best spatial homotopies from continuous splits;
 foreach ranked spatial homotopy **do**
 Solve path QP; map the full path envelope to robust ST conflicts;
 Generate causal temporal homotopies;
 foreach ranked temporal homotopy **do**
 Compile bidirectional ST bounds and solve speed QP;
 Store every robustly verified feasible trajectory;
 end
 end
 Stop if feasible or refinement is not triggered; otherwise update (15);
end
 Return the minimum-cost feasible trajectory and persist its semantic label;
Algorithm 1: MIKU Homotopy-Corridor Planning

Proposition 2 (Robust collision exclusion). *If $O_i^{\text{true}}(t) \subseteq O_i^{\text{rob}}(t)$ and the continuously executed trajectory remains in the corridor constructed by excluding O_i^{rob} and the vehicle footprint, it cannot intersect $O_i^{\text{true}}(t)$.*

Proof. Disjointness from a superset implies disjointness from every subset. $\square$

D. On-Demand Refinement and Receding-Horizon Consistency

The first arrival map follows a second-order kinematic estimate. When a purely dynamic conflict makes the first pass infeasible, the speed solution produces a first-arrival inverse $\hat{\tau}^{(r)}(s)$ and MIKU updates

$$\tau^{(r+1)}(s) = (1 - \alpha)\tau^{(r)}(s) + \alpha\hat{\tau}^{(r)}(s), \quad \alpha = 0.7. \quad (15)$$

At most two planning passes are performed, and the best verified feasible iterate is retained. Thus refinement repairs temporal mismatch on demand rather than adding unconditional cycles.

Across receding-horizon cycles, spatial and temporal choices carry semantic labels. Switching labels incurs a persistence cost. Moreover, after selecting a verified pass-before trajectory, MIKU commits to that trajectory until the ego clears the conflict. This prevents an unsafe late switch to yielding after the stopping line has already been crossed.

IV. EXPERIMENTAL EVALUATION

A. Protocol

We compare four methods on the same Apollo-style path-speed chain. B0 uses time-blind static path bounds, a fixed margin, and greedy obstacle-wise nudging. B1 adds arrival-time projection but retains greedy spatial decisions. B2 alternates path and speed timing for up to three iterations. MIKU enables robust occupancy, threat-adaptive margins, grouping, exact K-best spatial homotopies, the temporal graph, bidirectional ST bounds, and on-demand refinement. All methods share scenarios, predictions, vehicle geometry, QPs, dynamics, horizons, and objective weights.

Seven scenario families cover pedestrian crossing, vehicle cut-in, parked vehicle with oncoming traffic, narrow multi-obstacle passages, interleaved dynamic obstacles, noisy prediction, and delayed crossing. Each family uses 500 fixed seeds, yielding 3500 paired cases. The noisy family perturbs planner

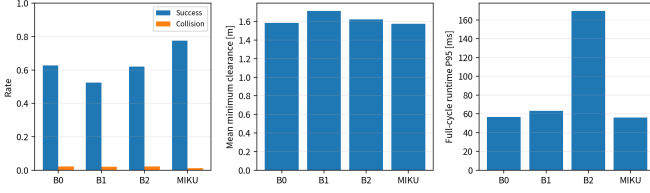


Fig. 2. Aggregate paired-randomized evaluation.

inputs while evaluation uses unperturbed truth. A success is a collision-free arrival at the evaluation line within the horizon. Collision is continuous-time overlap of physical ego/obstacle rectangles. Full-cycle latency includes constraint construction, candidate generation, both QPs, and triggered refinements.

Binary outcomes use an exact McNemar test. Percentage-point differences and continuous effects use 5,000 paired bootstrap resamples for 95% confidence intervals (CIs). The main timing experiment runs methods sequentially in one process on an Intel Core i9-14900HX under Linux 7.2.2 with OSQP 1.1.1 [30]. Raw cases, failures, statistics, plots, and manuscript macros are produced by one evaluation entry point.

B. Overall Performance

MIKU reaches 77.5% collision-free success versus 62.7%, 52.4%, and 62.1% for B0–B2 (Table II). The paired MIKU–B0 gain is 14.8 points (95% CI 13.5–16.0; exact McNemar $p = 1.939e - 133$). Collision falls from 2.2% to 1.1%, a paired difference of -1.0 points (95% CI -1.4–0.7; $p = 2.91e - 11$). The increased traversal capability therefore does not trade against safety.

MIKU’s P95 complete-planning latency is 55.93 ms, comparable to B0 at 56.79 ms and substantially below the unconditional iterative B2 at 169.49 ms. Its higher P99 reflects on-demand candidate fallback, while the P95 shows that most cases terminate on an early feasible homotopy.

C. Scenario-Wise Effects

The largest spatial gain occurs in narrow multi-obstacle passages: 65.4 points, consistent with continuous partitions recovering corridors discarded by greedy nudging. Delayed crossing improves by 26.8 points, demonstrating the value of explicit pass-before/yield-after choices. Under prediction noise, success improves by 3.2 points and collision changes by -7.0 points, directly supporting robust-tube propagation. Crossing and interleaved cases are already near saturation for B0, and MIKU maintains their success.

D. Large-Scale Paired Ablation

Each ablation uses the same seven families and 3500 paired seeds. A1 removes arrival-time initialization, A2 grouping, A3 K-best spatial homotopy, A4 threat-adaptive margins, A5 the temporal graph, A6 robust occupancy, and A7 alternating refinement. Differences in Table IV are full MIKU minus the ablation.

K-best spatial homotopy and threat-adaptive margins contribute 10.2 and 10.7 points, respectively. The temporal graph

adds 3.3 points. Removing robust occupancy increases collision from 1.1% to 2.2%; the paired collision difference is -1.1 points ($p = 1.455e - 11$). Refinement adds 0.8 points. A1 is not significant, indicating that kinematic arrival time is primarily an initializer; the final benefit arises from homotopy search, QP validation, and feedback refinement.

E. Receding-Horizon and Joint-Reference Evaluation

The receding-horizon protocol repeatedly plans, executes 0.5 s, updates ego and obstacles, and replans. It contains 100 seeds per family, or 700 paired episodes. MIKU reaches 72.3% versus B0’s 62.3%, a 10.0-point gain (95% CI 7.6–12.6; $p = 1.524e - 15$). Both collision rates are 1.3%. Semantic persistence and pass-before commitment therefore carry the open-loop benefit into repeated replanning without increasing collision.

The B3 reference jointly searches $(t, s, l, v, \dot{l})$ on a coarse grid and uses the same threat margins and robust prediction tubes as MIKU. It is a quality–cost reference, not a continuous global optimum. Across 70 paired cases, B3 and MIKU achieve 80.0% and 77.1%; the MIKU–B3 difference is -2.9 points (95% CI -11.4–5.7; $p = 0.7539$). Thus success is statistically indistinguishable at this sample size. B3’s P95 is 13363.83 ms versus 44.86 ms for MIKU, showing that finite homotopies plus convex QPs recover comparable feasibility at over two orders of magnitude lower computation. B3 has lower penalized travel time among its solutions, reflecting the remaining efficiency advantage of joint search.

F. Sensitivity and Scope

Perturbing all five threat weights by $\pm 20\%$ and renormalizing changes the resulting margin by at most 6.4%. Twenty full-planning perturbations in each of four representative scenarios (80 trajectories total) all succeed; the maximum lateral trajectory deviation is 0.011 m. The core effect is therefore locally stable rather than dependent on one exact weight vector.

The evaluation covers planner-in-the-loop structured-road scenarios and bounded prediction perturbations. It does not yet include perception false detections, complex intersection rules, or physical actuator errors. These are the next interfaces for probabilistic prediction and vehicle experiments, while the reported theorem, convexity property, executable implementation, and paired simulation evidence remain directly reproducible.

V. CONCLUSION

MIKU introduces an interaction-aware spatiotemporal homotopy corridor around a mature decoupled path–speed solver. A continuous-partition theorem converts 2^k passing-side assignments into $k + 1$ exact lateral bands; an exact K-best layered search preserves spatial alternatives; and a causal temporal graph compiles pass-before/yield-after orders into bidirectional ST bounds. Robust occupancy, envelope-aware mapping, on-demand refinement, and semantic commitment complete the planning loop while every fixed-homotopy problem remains a convex QP.

TABLE II
AGGREGATE RESULTS OVER SEVEN RANDOMIZED FAMILIES ($n = 3500$ PER METHOD)

Method	Success (%)	Collision (%)	Stop (%)	Clearance (m)	Jerk RMS	P50 (ms)	P95 (ms)	P99 (ms)
B0	62.7	2.2	35.4	1.586	1.25	10.11	56.79	138.88
B1	52.4	2.1	45.8	1.712	1.31	11.37	63.24	169.73
B2	62.1	2.2	36.0	1.624	1.30	31.20	169.49	463.05
MIKU	77.5	1.1	21.5	1.575	1.13	9.98	55.93	221.80

TABLE III
MIKU VS. B0 SUCCESS BY SCENARIO FAMILY

Family	B0	MIKU	Diff.	95% CI
Crossing	99.4	99.4	0.0	[0.0,0.0]
Cut-in	85.6	85.6	0.0	[0.0,0.0]
Parked/oncoming	53.0	61.0	8.0	[5.4,10.8]
Narrow multi-obs.	1.6	67.0	65.4	[61.2,69.6]
Interleaved	100.0	100.0	0.0	[0.0,0.0]
Prediction noise	36.4	39.6	3.2	[1.4,5.2]
Delayed crossing	62.8	89.6	26.8	[22.6,31.0]

Across 3500 paired random cases, MIKU improves collision-free arrival by 14.8 points and reduces collision by 1.0 point relative to B0, with P95 full-planning latency of 55.93 ms. The 700-episode receding-horizon study preserves a 10.0-point arrival gain. These results establish MIKU as a practical route to richer spatiotemporal decisions without replacing an existing path-velocity optimization stack.

REFERENCES

- [1] K. Kant and S. W. Zucker, "Toward efficient trajectory planning: The path-velocity decomposition," *The International Journal of Robotics Research*, vol. 5, no. 3, pp. 72–89, 1986. DOI: 10.1177/027836498600500304
- [2] M. Werling, J. Ziegler, S. Kammel, and S. Thrun, "Optimal trajectory generation for dynamic street scenarios in a Frenet frame," in *2010 IEEE International Conference on Robotics and Automation*, 2010, pp. 987–993. DOI: 10.1109/ROBOT.2010.5509799
- [3] D. Dolgov, S. Thrun, M. Montemerlo, and J. Diebel, "Path planning for autonomous vehicles in unknown semi-structured environments," *The International Journal of Robotics Research*, vol. 29, no. 5, pp. 485–501, 2010. DOI: 10.1177/0278364909359210
- [4] M. McNaughton, C. Urmson, J. M. Dolan, and J.-W. Lee, "Motion planning for autonomous driving with a conformal spatiotemporal lattice," in *2011 IEEE International Conference on Robotics and Automation*, 2011, pp. 4889–4895. DOI: 10.1109/ICRA.2011.5980223
- [5] J. Ziegler, P. Bender, M. Schreiber, et al., "Making Bertha drive—an autonomous journey on a historic route," *IEEE Intelligent Transportation Systems Magazine*, vol. 6, no. 2, pp. 8–20, 2014. DOI: 10.1109/MITS.2014.2306552
- [6] H. Fan et al., *Baidu Apollo EM motion planner*, 2018. arXiv: 1807.08048.
- [7] Apollo Auto. "Apollo open-source autonomous driving platform," Accessed: Sep. 3, 2026. [Online]. Available: <https://github.com/ApolloAuto/apollo>
- [8] Y. Zhang, H. Sun, J. Zhou, J. Pan, J. Hu, and J. Miao, "Optimal vehicle path planning using quadratic optimization for Baidu Apollo open platform," in *2020 IEEE Intelligent Vehicles Symposium*, 2020, pp. 978–984. DOI: 10.1109/IV47402.2020.9304787
- [9] J. Zhou et al., "Autonomous driving trajectory optimization with dual-loop iterative anchoring path smoothing and piecewise-jerk speed optimization," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 439–446, 2021. DOI: 10.1109/LRA.2020.3045925
- [10] M. Morsali, E. Frisk, and J. Åslund, "Spatio-temporal planning in multi-vehicle scenarios for autonomous vehicle using support vector machines," *IEEE Transactions on Intelligent Vehicles*, vol. 6, no. 4, pp. 611–621, 2021. DOI: 10.1109/TIV.2020.3042087
- [11] J. Hu et al., "Spatiotemporal joint planning for intelligent vehicles based on improved hybrid A*," *Automotive Engineering*, vol. 45, no. 7, pp. 1135–1147, 2023, (in Chinese). DOI: 10.19562/j.chinasae.qcgc.2023.07.003
- [12] F. Micheli, M. Bersani, S. Arrigoni, F. Braghin, and F. Cheli, "NMPC trajectory planner for urban autonomous driving," *Vehicle System Dynamics*, vol. 61, no. 5, pp. 1387–1409, 2023. DOI: 10.1080/00423114.2022.2081220
- [13] Z. Han et al., "An efficient spatial-temporal trajectory planner for autonomous vehicles in unstructured environments," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 2, pp. 1797–1814, 2024. DOI: 10.1109/TITS.2023.3315320
- [14] J. Guo et al., "Spatio-temporal joint optimization-based trajectory planning method for autonomous vehicles in complex urban environments," *Sensors*, vol. 24, no. 14, p. 4685, 2024. DOI: 10.3390/s24144685
- [15] J. Hu, J. Zheng, S. Zhou, W. Zhao, Z. Zhang, and M. Yao, "Spatiotemporal joint trajectory planning for intelligent vehicles on structured roads," *Automotive Engineering*, vol. 47, no. 5, 2025, (in Chinese). DOI: 10.19562/j.chinasae.qcgc.2025.05.003
- [16] J. Chen, W. Zhan, and M. Tomizuka, "Autonomous driving motion planning with constrained iterative LQR," *IEEE Transactions on Intelligent Vehicles*, vol. 4, no. 2, pp. 244–254, 2019. DOI: 10.1109/TIV.2019.2904385
- [17] J. Cheng, Y. Chen, Q. Zhang, L. Gan, C. Liu, and M. Liu, "Real-time trajectory planning for autonomous driving with gaussian process and incremental refinement," in *2022 International Conference on Robotics*

TABLE IV
PAIRED COMPONENT ABLATION OVER 3500CASES

Configuration	Success (%)	Collision (%)	Gain (points)	95% CI	McNemar p
Full MIKU	77.5	1.1	—	—	—
A1 w/o arrival initialization	77.3	1.2	0.1	[-0.1,0.4]	0.5235
A2 w/o grouping	76.9	1.1	0.5	[0.3,0.8]	7.629e-06
A3 w/o K-best spatial homotopy	67.3	1.1	10.2	[9.2,11.2]	1.363e-107
A4 w/o threat margin	66.7	1.1	10.7	[9.7,11.8]	1.299e-113
A5 w/o temporal graph	74.2	1.2	3.3	[2.5,4.0]	1.342e-18
A6 w/o robust occupancy	77.0	2.2	0.5	[0.2,0.8]	0.001514
A7 w/o refinement	76.6	1.0	0.8	[0.5,1.1]	3.725e-09

TABLE V
RECEDING-HORIZON AND JOINT-GRID PROTOCOLS

Protocol	Method	Success	Collision	P95 ms
Rolling ($n = 700$)	B0	62.3	1.3	827.20
	MIKU	72.3	1.3	1361.51
Joint ($n = 70$)	B3	80.0	1.4	13363.83
	MIKU	77.1	0.0	44.86

and Automation, 2022, pp. 8999–9005. DOI: 10.1109/ICRA46639.2022.9812405

- [18] Z. Zhang, C. Wang, W. Zhao, M. Cao, J. Liu, and K. Xu, “Path-speed decoupling planning method based on risk cooperative game for intelligent vehicles,” *IEEE Transactions on Transportation Electrification*, vol. 10, no. 1, 2024. DOI: 10.1109/TTE.2023.3316124
- [19] Y. Yoon, C. Kim, H. Lee, D. Seo, and K. Yi, “Spatio-temporal corridor-based motion planning of lane change maneuver for autonomous driving in multi-vehicle traffic,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 10, pp. 13 163–13 183, 2024. DOI: 10.1109/TITS.2024.3388380
- [20] F. M. Tariq, Z.-H. Yeh, A. Singh, D. Isele, and S. Bae, “Frenet corridor planner: An optimal local path planning framework for autonomous driving,” in *2025 IEEE Intelligent Vehicles Symposium*, 2025, pp. 686–693. DOI: 10.1109/IV64158.2025.11097649
- [21] F. M. Tariq, Z.-H. Yeh, A. Singh, D. Isele, and S. Bae, “Frenet enveloping planner: An efficient local path planning framework for autonomous driving,” *IEEE Transactions on Intelligent Vehicles*, vol. 11, no. 3, pp. 361–370, 2026. DOI: 10.1109/TIV.2025.3623575
- [22] Y. Kuwata, J. Teo, S. Karaman, E. Frazzoli, G. Fiore, and J. P. How, “Real-time motion planning with applications to autonomous urban driving,” *IEEE Transactions on Control Systems Technology*, vol. 17, no. 5, pp. 1105–1118, 2009. DOI: 10.1109/TCST.2008.2012116
- [23] T. Kessler, K. Esterle, and A. Knoll, “Mixed-integer motion planning on German roads within the Apollo driving stack,” *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 1, pp. 851–867, 2023. DOI: 10.1109/TIV.2022.3162671
- [24] B. Xu, S. Yuan, X. Lin, M. Hu, Y. Bian, and Z. Qin, “Space discretization-based optimal trajectory planning

for automated vehicles in narrow corridor scenes,” *Electronics*, vol. 11, no. 24, p. 4239, 2022. DOI: 10.3390/electronics11244239

- [25] W. Wu, H. Jia, Q. Luo, and Z. Wang, “Dynamic path planning for autonomous driving on branch streets with crossing pedestrian avoidance guidance,” *IEEE Access*, vol. 7, pp. 144 720–144 731, 2019. DOI: 10.1109/ACCESS.2019.2938232
- [26] B. Yang, S. Yan, Z. Wang, and K. Nakano, “Prediction-based trajectory planning for safe interactions between autonomous vehicles and moving pedestrians in shared spaces,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 10, pp. 10 513–10 524, 2023. DOI: 10.1109/TITS.2023.3281157
- [27] F. Liu, X. Zhao, W. Su, and Y. Wen, “Dynamic path-speed planning algorithm for autonomous driving on structured roads,” *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, vol. 238, no. 10–11, pp. 3172–3193, 2024. DOI: 10.1177/09544070231181626
- [28] Y. Hu et al., “Planning-oriented autonomous driving,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 17 853–17 862.
- [29] B. Jiang et al., “VAD: Vectorized scene representation for efficient autonomous driving,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 8340–8350.
- [30] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, “OSQP: An operator splitting solver for quadratic programs,” *Mathematical Programming Computation*, vol. 12, no. 4, pp. 637–672, 2020. DOI: 10.1007/s12532-020-00179-2